



Indian Institute of Technology Kanpur
Department of Mathematics and Statistics
An Introduction To Bayesian Analysis (MTH422A)
End Semester Examination
Date: April 24, 2024, Wednesday

Timing: 12:30 AM – 2:30 PM (2 hours)

Spring 2024

Max point: 25

The Weibull distribution has a density function $f(x) = \alpha\beta x^{\alpha-1} \exp[-\beta x^\alpha]$ for $x \in \mathbb{R}^+$. Here the shape and rate parameters are given by $\alpha > 0$ and $\beta > 0$. Suppose $X_1, \dots, X_n$ denote the annual total rainfall at IITK Airstrip for n years. You model the data as IID observations (assuming “our tolerance and habits have changed but the climate didn’t”) from a Weibull distribution with known shape $\alpha = \alpha_0$ (unless explicitly mentioned that it is unknown as in Q7) but assume that β is unknown. If you cannot solve Q1, then I will tell you the answer. In that case your marks in Q1 will be zero but it will save marks in other questions. If answer to Q1 is wrong, the most of the other answers will be wrong and so be careful. In Q7 and Q8, you have to explicitly mention the steps like initialization, updates, discarding undesired samples, etc. up to the information asked in the questions. For each question where the answer is a single number or expression, put a box around your answer to highlight.

1. Choose a conjugate prior for β with generic choice of hyperparameters (i.e., if it is a normal distribution, choose its mean and variance to be some m and s^2 but don't fix $m = 0, s = 100$). Show that your choice is indeed a conjugate prior. No point will be given if you don't show. (2 points)
2. Calculate the posterior mean, variance, 95% equal-tailed exact credible interval, and 95% Gaussian-approximated asymptotic credible interval for β . (1+1+1+1=4 points)
3. Prediction of the amount of rainfall for the $(n+1)$ -th year is necessary for agricultural experiments going on at IITK. Here, not only the predicted value is important but also the whole predictive distribution is important for accurate uncertainty quantification. Derive the analytical expression of the posterior predictive density function. (2 points)
4. We want to decide between two models

$$\mathcal{M}_1 : X_i \stackrel{\text{iid}}{\sim} \text{Weibull}(\alpha_0, \beta_0), \quad \mathcal{M}_2 : X_i | \beta \stackrel{\text{iid}}{\sim} \text{Weibull}(\alpha_0, \beta), \quad \beta \sim \pi(\beta),$$

where $\pi(\beta)$ is as in Q1. Find the Bayes factor $BF_{2,1}$ in terms of $n, \alpha_0, \beta_0, X_i$'s and the hyperparameters in $\pi(\beta)$. (2 points)

5. Two faculties from the Department of Earth Sciences are unhappy about the non-uniqueness of the final inferences due to the choice of priors. Profs. A and B want to choose priors $\pi_A(g(\beta))$ and $\pi_B(h(\beta))$ for two different transformations of β given by $g(\beta)$ and $h(\beta)$ respectively, but they want to ensure that both of them end up with the same posterior for β . Derive a prior for β that ensures this condition. (2 points)
6. The mean of a $\text{Weibull}(\alpha_0, \beta)$ distributed random variable is given by $\Gamma(1 + \alpha_0^{-1})\beta^{-1}$. Hence, we are interested in summarizing the posterior of β through a single measure that minimizes the posterior risk under the loss function $\ell(\hat{\beta}, \beta) = [\hat{\beta}^{-1} - \beta^{-1}]^2$. Find the Bayes rule. (2 points)

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7. In practice, we never know α_0 and so one option is to treat α as unknown along with β . Here no known conjugate prior exists for α . To draw Bayesian inference, we now need to draw samples from the posterior distribution $\pi(\alpha, \beta | X_1, \dots, X_n)$. Write down a Metropolis-Hastings within Gibbs sampling scheme here. How will you use these samples to further approximate the posterior predictive density? (3+1 = 4 points)
8. Prof. C from the Department of Earth Sciences does not believe that the climate didn't change and hence he prefers to check whether there is any trend in the data or not. However, due to his limited knowledge in generalized linear models, he transform the data into the log-scale $Y_i = \log[X_i]$, $i = 1, \dots, n$ and fit a model $Y_i | \beta_0, \beta_1, \sigma^2 \stackrel{\text{Indep}}{\sim} \text{Normal}(\beta_0 + \beta_1 Z_i, \sigma^2)$, where Z_i 's are some known transformed forms of $i = 1, \dots, n$. Hence, if β_1 is significantly away from zero, we would conclude that there is a trend in the data across years. He wants to perform Bayesian LASSO. Write a Gibbs sampling algorithm that does not involve any Metropolis-Hastings step or any accept-reject-type step. Choose priors properly and mention the full conditional distributions of parameters and hyperparameters. You don't have to derive the parameters of the full conditional distributions explicitly but just mention the names of those distributions only. (3 points)
9. Prof. C found that zero lies within the 95% credible interval of β_1 and hence he concludes that there is no significant linear trend in log-rainfall. You are not convinced because Bayesian LASSO shrinks the posterior distribution of β_1 towards zero. You want to compare the results with a simple Bayesian regression model with noninformative priors. Mention four ways (just the names) of doing this. (2 points)
10. Unobservable climate conditions can possibly induce dependence in the observations. Hence, Prof. E from the Department of Computer Science and Engineering wants to explore a hidden Markov model as follows:

$$Y_i | W_i, \beta_0, \beta_1, \sigma^2 \stackrel{\text{Indep}}{\sim} \text{Normal}(\beta_0 + \beta_1 Z_i + W_i, \sigma^2)$$

$$W_1 \sim \text{Normal}(0, \tau^2), W_i | W_{i-1} \sim \text{Normal}(\rho W_{i-1}, \tau^2(1 - \rho^2)), i = 2, \dots, n$$

$$\beta_0, \beta_1 \stackrel{\text{iid}}{\sim} \text{Normal}(0, 100^2), \sigma^2, \tau^2 \stackrel{\text{iid}}{\sim} \text{Inverse-gamma}(0.01, 0.01), \rho \sim \text{Uniform}(-1, 1)$$

Write down a directed acyclic graph (DAG) representation of the above model. (2 points)